

🔗 30-Second Elevator Pitch for Judges:

"Autonomous vehicles today blindly trust their sensors. If an attacker blinds a Tesla camera with a laser or spoofs an ultrasonic sensor with acoustic noise, the car crashes. **TrustEdge** introduces a mathematical Trust Engine that runs directly at the hardware edge. It cross-arbitrates between different branches of physics (light vs. sound) and checks every reading against Newtonian mechanics. If any sensor lies, our 4-Stage Hardware Safety FSM automatically throttles speed or cuts motor power in silicon before an accident occurs."

📐 1. The Actual Mathematics (Explained in Simple Terms)

The system evaluates sensor integrity every **100 milliseconds** ($\Delta t = 0.1 \text{ s}$) across three complementary physical pillars:

Pillar 1: Cross-Modal Spatial Arbitration (Light vs. Sound)

P1 PENALTY = 70 (CRASH THREAT) / 25 (GHOST WALL)

Physical Principle: Photons (Optical IR) travel at $300,000 \text{ km/s}$; acoustic waves (HC-SR04) travel at 343 m/s . When an obstacle is directly in front of the vehicle, both sensors *must* agree.

Blindspot Attack: If $(IR1 = 1 \text{ OR } IR2 = 1) \text{ AND } (Distance_US > 35.0 \text{ cm}) \rightarrow \text{Penalty P1} = 70$

Ghost Wall: If $(Distance_US < 15.0 \text{ cm}) \text{ AND } (IR1 = 0 \text{ AND } IR2 = 0) \rightarrow \text{Penalty P1} = 25$

Why it catches attacks: If an acoustic absorber hides a wall, ultrasound claims "145 cm clear", but optical IR physically senses the barrier at 3 cm $\rightarrow$ immediate emergency brake!

Pillar 2: Kinematic Velocity Bounding (Newtonian Mechanics)

P2 PENALTY = 30 (KINEMATIC VIOLATION)

Physical Principle: Teleportation is physically impossible. With maximum rover speed $v_{\max} = 0.45 \text{ m/s}$, an obstacle cannot jump more than 75 cm closer or farther in a tenth of a second.

Apparent Velocity: $v_{\text{apparent}} = |Distance_k - Distance_{(k-1)}| / \Delta t$
If $|Distance_k - Distance_{(k-1)}| > 75.0 \text{ cm}$ ($> 27 \text{ km/h}$) $\rightarrow \text{Penalty P2} = 30$

Why it catches attacks: Attackers injecting pulsed acoustic jitter cause readings to teleport (12 cm $\rightarrow$ 140 cm). The kinematic gate trips instantly and throttles motor speed.

Pillar 3: Sensor Stagnation & Micro-Variance (Statistical Entropy)

P3 PENALTY = 46 (SENSOR REPLAY / FROZEN)

Physical Principle: A moving vehicle naturally has chassis vibrations and road noise. A genuine acoustic sensor *never* produces the exact same float across 10 cycles. Zero variance proves artificial replay.

Variance: $\sigma^2 = (1/N) \sum (d_i - d_{\text{mean}})^2$ over N=10 frames
If Motors_Active AND $\sigma^2 < 0.0001 \text{ cm}^2 \rightarrow \text{Penalty P3} = 46$

Why it catches attacks: If an attacker replays a stale packet stream ("42.000 cm"), the zero-entropy detector detects stagnation and forces Limp-Home mode.

Persistence Escalator (μ) & Asymmetric EMA Smoothing

DYNAMIC SAFETY HYSTERESIS

Single Glitch vs. Persistent Attack: A single bad packet shouldn't stall the vehicle. A memory integrator increments on bad frames ($C_k \leq 5$). Total penalty: $P_{\text{total}} = P_{\text{raw}} \cdot (1.0 + 0.03 \cdot C_k)$.

Trust Score: $T_k = (1 - \alpha) \cdot T_{(k-1)} + \alpha \cdot T_{\text{target}}$
 $\alpha_{\text{drop}} = 0.65$ (Fast Attack Reaction: 80 ms) | $\alpha_{\text{rise}} = 0.12$ (Cautious Recovery: 800 ms)

🚦 2. The 4-Stage Hardware Safety FSM

The continuous mathematical Trust Score $T \in [0, 100]$ maps deterministically into 4 discrete hardware operational regimes enforced directly in the Arduino Uno firmware registers:

Safety State	Trust Score	Hardware PWM	Authority	Vehicle Behavior & Safety Response
NOMINAL	80.0% – 100%	255 PWM	100% Power	All sensors physically consistent. Full cruising speed authorized.
DEGRADED	60.0% – 79.9%	140 PWM	55% Power	Non-fatal anomaly (Ghost Wall / Jitter). Speed throttled for caution.
LIMP_HOME	40.0% – 59.9%	85 PWM	33% Power	Primary sensor failed (Freeze). Crawl speed; status LED blinks at 2 Hz.
SAFE_STOP	0.0% – 39.9%	0 PWM	0% Power	Critical crash threat (Blindspot). Motors hard-locked in silicon; latched stop.

⚡ 3. The Cyber-Physical Attack Injection Deck

Each button in the Cockpit demonstrates a distinct level of the defense hierarchy:

Attack Button	Simulated Injection	Score	Triggered State	Judge Explanation
⚠️ Blindspot Spoof	US=145cm, IR1=1	0.0%	SAFE_STOP (0 PWM)	Optical detects barrier right in front, ultrasound lied. Instant emergency brake.
👻 Ghost Wall	US=5cm, IRs=0	~71.3%	DEGRADED (140 PWM)	Phantom wall reported. Safe to drive cautiously at 55% speed; does not freeze.
⚡ Kinematic Jitter	US 12cm ↔ 140cm	~65.5%	DEGRADED (140 PWM)	Acoustic multipath reflections detected. Speed clamped to safe handling ceiling.
🧊 Replay Freeze	US=42.000cm (0 var)	~47.1%	LIMP_HOME (85 PWM)	Primary acoustic ranger frozen. Rover creeps home on secondary optical bumpers.
🔧 Restore Hardware	Live Sensor Bus	100.0%	NOMINAL (255 PWM)	One-click unlatch: resets persistence, re-arms motors to 100% full authority.

🔧 4. Physical Hardware Architecture & Firmware Innovations

Microcontroller & Motor Drivers

- **Arduino Uno (ATmega328P @ 16 MHz):** Executes non-blocking serial parsing in $<1\ \mu\text{s}$.
- **HW-130 / L293D Shield:** 74HC595 shift register controls direction; Timer 0 PWM controls speed on Pins 5 & 6.
- **Hardware Speed Clamping:** $u_{\text{eff}} = \min(\text{speed}, \text{fsmMaxPwm})$ enforced in firmware silicon.

Communications & Servo Innovations

- **Tower Pro MG90S Auto-Detach:** Firmware detaches servo 300ms after move → eliminates 100% of timer jitter and buzzing!
- **HC-05 Bluetooth (9600 baud):** Direct 2-byte protocol ("W\n", "X\n") executes in $<2\text{ms}$ without buffer overflows.
- **Zero-Lag Asymmetric DSP:** Brakes instantaneously on approach (0ms lag), filters noise in open space.

5. "Where is the Proof?" — Live Demonstration Script

Perform these physical demonstrations to prove the system is running real math, not pre-programmed animations:

- 1 The Cross-Modal Physical Bumper Proof:** Point the rover into open air (Cockpit reads 150cm, NOMINAL 100%). Press **W** to start driving forward. Now put your finger 2cm in front of the Left Optical IR bumper (Pin A0) while the ultrasonic sensor points into open space. *Result:* Light detects the obstacle, ultrasound says clear road. The cross-modal engine trips Pillar 1, drops trust to 0%, and locks the motors immediately!
- 2 The Graduated FSM State Transition:** Click  **Ghost Wall** → watch the cockpit turn yellow (DEGRADED, 140 PWM) and observe the rover driving at 55% speed. Click  **Replay Freeze** → watch the cockpit turn orange (LIMP-HOME, 85 PWM) and the onboard LED start blinking. Click  **Blindspot Spoof** → watch the red alarm flash and the brakes hard-lock. Click **Restore Hardware Stream** → watch it snap instantly back to 100% NOMINAL!
- 3 Firmware Obstacle Safety Interlock:** Drive the rover toward a real physical wall. Because of the Asymmetric Zero-Lag Filter ($\text{dist} < \text{smoothedDistance}$), the rover senses the closing wall and stops with a safe 8cm buffer without colliding!

6. Top 5 Tough Judge Questions & Winning Answers

Q1: "Why do you need both Optical IR and Ultrasonic sensors? Isn't one enough?"

Answer: "No single sensor modality is secure. Ultrasonic sensors can be jammed with cheap speakers or sound-absorbing acoustic foam. Optical sensors can be blinded by sunlight or clear glass. By cross-arbitrating light waves against sound waves, an attacker must simultaneously spoof two completely separate branches of physics at the exact same microsecond."

Q2: "Why didn't you just use a standard Kalman Filter?"

Answer: "A Kalman Filter assumes Gaussian noise (random bell curve). But cyber-physical spoofing is malicious and non-Gaussian. If an attacker injects a lie, a Kalman Filter averages the lie into the truth, leading to catastrophic vehicle drift. TrustEdge uses deterministic physics gating: if a reading violates Newtonian mechanics, it is rejected with zero trust."

Q3: "What happens if Bluetooth disconnects while the rover is moving?"

Answer: "The Arduino firmware runs a dedicated hardware fail-safe watchdog (WATCHDOG_TIMEOUT_MS = 3000). If no valid command packet is received within 3 seconds, the microcontroller automatically cuts motor power. The car will never run away."

Q4: "How does this prevent the Tower Pro servo from buzzing or vibrating?"

Answer: "Over Bluetooth, SoftwareSerial disables global interrupts during transmission, which causes timing jitter on Timer 1 and shakes the servo. We engineered a 300ms Auto-Detach Engine: once the servo reaches target angle, the firmware automatically detaches it, shutting off PWM pulses and completely eliminating jitter and power drain."

Q5: "How does this scale up to real autonomous vehicles like Tesla or Waymo?"

Answer: "The exact same 3 pillars scale directly. In a full-size vehicle, Pillar 1 arbitrates LiDAR (light) against Radar (radio) and Cameras (photons). Pillar 2 enforces vehicle kinematics ($v \leq 130 \text{ km/h}$, $a \leq 1g$). Pillar 3 detects frozen CAN bus packets. Our mathematical model provides the verifiable safety envelope that sits between AI perception and drive-by-wire actuators."